Integration Testing Report

StreetRace Manager

Part 2: Integration Testing

Test Framework:	pytest (Python)
Additional Modules:	Finance, Leaderboard
Test Files:	5 integration test files
All Tests:	Pass

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1. Additional Modules Designed

As part of the system requirements, two additional modules were designed to extend the core functionality of the StreetRace Manager:

1. **Finance Module** — Manages the team's income and expenses. It records race prize money as income and handles any expenses. It ensures all financial transactions are logged and tracks the total balance.
2. **Leaderboard Module** — Tracks the total wins of drivers and ranks them accordingly. It updates automatically when race results are recorded and allows retrieving the top drivers or a specific driver's position.

2.1 Call Graph

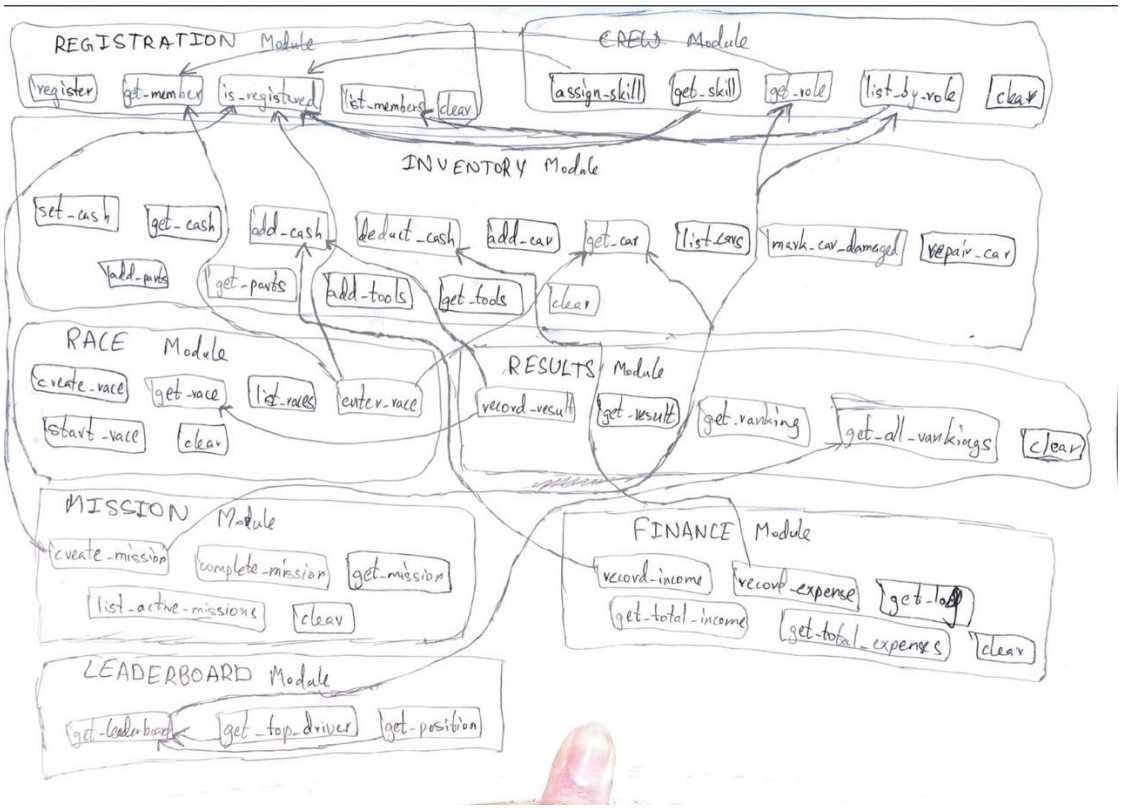

Figure 1: Call Graph — StreetRace Manager Integration

2.2 Integration Test Design

Below is the design of integration test cases validating how different modules interact with one another.

Scenarios Tested in `test_integration_registration-race.py`

Test Case: `test_registered_driver_can_enter_race`

Field	Detail
Modules Involved	Registration, Crew Management, Race Management
Expected Result	A successfully registered driver can be added to a race.
Actual Result	**Pass**
Why it is needed	Verifies the core flow of registering a crew member, assigning a driver role, and successfully entering them in a race.

Test Cases: Rejection of invalid entrants

`test_unregistered_person_cannot_enter_race`, `test_mechanic_cannot_enter_race`, `test_strategist_cannot_enter_race`

Field	Detail
Modules Involved	Registration, Crew Management, Race Management
Expected Result	System rejects unregistered users or non-drivers (mechanics/strategists) from entering races.
Actual Result	**Pass**
Why it is needed	Enforces the business rule that *only* registered members with the "driver" role can enter a race.

Scenarios Tested in `test_integration_damaged.py`

Test Cases: Car damage and race eligibility

`test_damaged_car_cannot_enter_race`, `test_repaired_car_can_enter_race`

Field	Detail
Modules Involved	Inventory, Race Management
Expected Result	A race cannot start with a damaged car, but it can if the car is repaired.
Actual Result	**Pass**
Why it is needed	Verifies the interaction between car condition (Inventory) and race entry (Race Management).

Test Cases: Damaged car and mission planning

`test_damaged_car_mission_blocked_no_free_mechanic`, `test_damaged_car_mission_allowed_mechanic_is_free`

Field	Detail
Modules Involved	Inventory, Crew Management, Mission Planning
Expected Result	A damaged car triggers a repair mission only if a mechanic is available.
Actual Result	**Pass**

Field	Detail
Why it is needed	Enforces the rule that missions requiring a mechanic must check for role availability.

Scenarios Tested in `test_integration_results.py`

Test Cases: Race completion effects

`test_completing_race_updates_cash`, `test_completing_race_updates_driver_ranking`

Field	Detail
Modules Involved	Results, Inventory, Crew Management
Expected Result	Completing a race appropriately updates the cash balance and increases the driver's win count.
Actual Result	**Pass**
Why it is needed	Ensures that race outcomes propagate correctly to financial tracking and driver statistics.

Scenarios Tested in `test_integration_mission.py`

Test Cases: Mission role verification

`test_mission_requires_correct_role`, `test_mission_cannot_start_if_role_unavailable`, `test_unregistered_member_cannot_get_mission`

Field	Detail
Modules Involved	Mission Planning, Crew Management
Expected Result	Missions are only assigned to registered crew members with the correct and available role.
Actual Result	**Pass**
Why it is needed	Validates role verification and availability checks before assigning tasks.

Scenarios Tested in `test_integration_finance.py`

Test Cases: Financial transaction logging

`test_race_prize_logged_as_income`, `test_expense_reduces_inventory_cash`

Field	Detail
Modules Involved	Finance, Results, Inventory
Expected Result	Race prizes increase cash and log as income; expenses decrease cash and log correctly.

Field	Detail
Actual Result	**Pass**
Why it is needed	Validates the end-to-end flow of money from racing rewards to inventory balance.

Scenarios Tested in `test_integration_leaderboard.py`

Test Cases: Leaderboard accuracy

`test_leaderboard_updates_after_single_race`,
`test_leaderboard_sorted_most_wins_first`

Field	Detail
Modules Involved	Leaderboard, Results, Crew Management
Expected Result	The leaderboard accurately reflects and sorts driver rankings based on updated race results.
Actual Result	**Pass**
Why it is needed	Ensures the new Leaderboard extra module properly consumes result data to maintain a global ranking.